

# KUNAL AGRAWAL

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[Portfolio](#) — [LinkedIn](#) — [Google Scholar](#)

## PROFESSIONAL SUMMARY

Computer Science PhD researcher and AWS Certified Cloud Practitioner specializing in Computer Vision, Multi-modal Data Fusion, and Generative AI. Proven track record of publishing high-impact research (including two journal publications and five conference papers) and deploying scalable AI workflows for real-world applications including healthcare, autonomous monitoring, and financial infrastructure.

## EDUCATION

### Doctor of Philosophy in Computer Science

*University of Dayton, OH*

**Aug 2024 – May 2027 (Expected)**

GPA: 3.96

- **Research Focus:** Saliency Detection, Scene Graph Generation, Visual Working Memory, Multi-modal Data Fusion.
- **Awards:** Research and Scholarship Award (Doctoral Student).

### Masters of Science in Computer Science

*University of Dayton, OH*

**Jan 2022 – Dec 2023**

GPA: 4.00

- **Awards:** Outstanding Graduate Student in Computer Science Award.

### Bachelor of Engineering in Information Technology

*Shri Ramdeobaba College of Engineering and Management, India*

**May 2021**

GPA: 3.66

## RESEARCH & INDUSTRY EXPERIENCE

### Graduate Research Assistant (PhD)

*University of Dayton, OH*

**Aug 2024 – Present**

- Developed the SeMemo dataset using Generative AI to disentangle semantic content from low-level visual features for Visual Working Memory (VWM) analysis, evaluating human memory fidelity via a change detection paradigm (Under Review, ECCV 2026).
- Engineered a comprehensive multi-modal house price prediction framework integrating street-view visual features, interior attributes, and macro-economic data using XGBoost and Neural Networks.
- Spearheaded the development of the AEye system, a specialized framework for automated avian monitoring utilizing high-resolution streaming video data analysis.
- Researched object-level visual attention and saliency detection by leveraging Scene Graphs and large-scale vision-language models like Kosmos-2 for advanced scene understanding.

### AI in Action Job Simulation

*Vista Equity Partners (Remote)*

**Jan 2026**

- Applied Vista's four-part prompt engineering framework (Ask, Context, Examples, Desired Output) to synthesize customer NPS survey feedback into executive-ready insights, improving clarity and relevance of outputs by > 50%.
- Designed and tested repeatable GenAI workflow automations for news aggregation, data reconciliation, and qualitative insight summarization using ChatGPT and Copilot.
- Evaluated AI-generated outputs for accuracy and tone, implementing guardrails and refinement techniques that significantly reduced error rates and increased stakeholder usability.

### Graduate Research Assistant (Master's)

*University of Dayton, OH*

**Aug 2022 – Dec 2023**

- Directed high-impact computer vision systems, achieving 89% accuracy in traffic video anomaly detection, and developed robust models for cryptocurrency price prediction.
- Implemented advanced machine learning and data analysis algorithms for model optimization, boosting overall working efficiency by 20%.
- Compiled and processed a dataset of 10,000 annotated images and 5,000 hours of video footage to train real-time detection models.

### Research Intern

*Tan Tock Seng Hospital, Singapore*

**June 2022 – July 2022**

- Navigated healthcare data science analysis on three distinct datasets containing 2,000+ patients each, extracting valuable diagnostic insights.
- Developed an automated image segmentation tool and streamlined data processes using ImageJ (Fiji) and R, facilitating Granuloma detection across 60,000+ medical images.

## PUBLICATIONS

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### Under Review & To Appear:

1. **K. Agrawal**, B. T. Tran-Le, M. T. Tran, T. N. Le, and T. V. Nguyen, "Semantic Matters in Short-Term Image Memorability," *European Conference on Computer Vision (ECCV)*, 2026. [**Under Review**].
2. **K. Agrawal**, J. N. R. Merakanapalli, J. Zeng, L. Nguyen, T. N. Le, M. T. Tran, and T. V. Nguyen, "SALOCON: Salient Object Detection in Context," *ACM International Conference on Multimedia (ACM MM)*, 2026. [**Under Review**].
3. A. Yousif, **K. Agrawal**, B. T. Tran-Le, V. Patel, and T. V. Nguyen, "House Price Prediction via Attribute, Visual, and Economic Features," *International Symposium on Information and Communication Technology (SoICT)*, 2025. [**To Appear**].
4. K. Jamale, **K. Agrawal**, B. T. Tran-Le, J. Merakanapalli, S. Chousalkar, V. Patel, T. N. Le, and T. V. Nguyen, "AEye: Avian Monitoring from Streaming Videos," *International Symposium on Information and Communication Technology (SoICT)*, 2025. [**To Appear**].

### Published:

5. B. T. Tran-Le, V. Patel, V. T. Huynh, M. K. Tran, **K. Agrawal**, M. T. Tran, and T. V. Nguyen, "Towards safer roads: benchmarking object detection models in complex weather scenarios," *Machine Vision and Applications*, vol. 36, no. 4, 94, 2025.
6. **K. Agrawal**, V. S. Patel, I. Cannon, M. T. Tran, and T. V. Nguyen, "Motorcycle Helmet Detection Benchmarking," *International Symposium on Information and Communication Technology*, pp. 203–215, 2024.
7. **K. Agrawal**, V. S. Patel, R. Tharra, T. N. Le, M. T. Tran, and T. V. Nguyen, "Motion Analysis in Static Images," *International Symposium on Information and Communication Technology*, pp. 193–202, 2024.
8. V. S. Patel, **K. Agrawal**, S. S. Baraheem, A. Yousif, and T. V. Nguyen, "Image de-photobombing benchmark," *Multimedia Tools and Applications*, vol. 83, no. 29, pp. 73553–73568, 2024.
9. V. S. Patel, **K. Agrawal**, and T. V. Nguyen, "A comprehensive analysis of object detectors in adverse weather conditions," *2024 58th Annual Conference on Information Sciences and Systems (CISS)*, pp. 1–6, 2024.

## TEACHING & ACADEMIC LEADERSHIP

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### Graduate Teaching Assistant / Lead Instructor

Aug 2025 – Present

University of Dayton, OH

- Served as the Lead Instructor for hands-on Python laboratory sessions, delivering technical lectures to undergraduate and graduate Computer Science students.
- Designed course lab materials, managed student inquiries, and provided technical guidance for software implementation via office hours.

### Academic Service & Mentorship

Jan 2024 – Present

University of Dayton, OH

- Actively served as a technical reviewer for academic conferences in Computer Vision and Machine Learning, evaluating novel research submissions for methodology, rigor, and impact.
- Mentored junior graduate students within the vision lab, providing hands-on guidance with experimental design, algorithm implementation (PyTorch, OpenCV), and the academic publication pipeline.

## TECHNICAL SKILLS

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- **AI & Machine Learning:** Generative AI, Prompt Engineering, LLMs (Kosmos-2), Saliency Detection, Scene Graph Generation, Multi-modal Data Fusion, Deep Learning, XGBoost.
- **Computer Vision:** Image Memorability Analysis, Object Detection (Adverse Weather), Image Restoration, Motion Analysis, Seam Carving.
- **Programming Languages:** Python (PyTorch, TensorFlow, Keras, OpenCV), C, C++, MATLAB, Java, SQL, R, Kotlin.
- **Tools & Infrastructure:** AWS (EC2, S3, RDS), Linux, LaTeX, Android Studio, Git, Tableau, ImageJ (Fiji).

## CERTIFICATIONS

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- **AWS Certified Cloud Practitioner** — **Cisco Data Analytics Essentials** — **Microsoft Technology Associate (MTA): Cloud & Networking Fundamentals**